THEO CASTILLO

Computer Science Student, seeking an engineering apprenticeship
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PROFESSIONAL EXPERIENCE

AI Developer, Work-Study Program — Sept. 2025 – Present
Genial (AI startup), Bordeaux, France

- Designed and deployed specialized AI agents for B2B clients, from requirements analysis to production deployment
- Built and managed RAG pipelines with context engineering, multi-agent orchestration and MCP servers
- Containerized services with Docker, set up CI/CD pipelines for reliable deployments
- Managed iterative client feedback loops and continuously adapted deployed solutions

AI Agent Builder, Freelance — Jun. – Jul. 2025
Genial, Bordeaux, France

- Built and maintained specialized AI agents with full project ownership
- Created and managed a sole proprietorship (first entrepreneurial experience)

Machine Learning Research Intern — Apr. – Jun. 2025
Bekki Lab, Ochanomizu University, Tokyo, Japan

- Reproduced the GPT-2 architecture in Haskell using Hasktorch (functional programming)
- Worked in an NLP/ML research environment, exploring formal approaches to deep learning

EDUCATION

B.Sc. in Computer Science, international track — 2023 – 2026
IUT de Gradignan, University of Bordeaux
Ranked 1st out of 66 students (semester 5). Highest recommendation from teaching staff.

Technological Baccalaureate (STI2D), with highest honors — 2020 – 2023
Lycee Gustave Eiffel

TECHNICAL PROJECTS (tcastillo.me)

VeridisQuo, deepfake detection [Python · PyTorch · YOLO · OpenCV · FastAPI]
Deepfake detection app combining spatial and frequency analysis. Face detection with YOLO, PyTorch classifier, real-time visualization and FastAPI backend.

Neural network library from scratch [Python · NumPy]
Full implementation in NumPy: linear layers, activation functions, loss functions, optimizers and backpropagation.

DeepRubiks, solving Rubik's Cube with deep learning [PyTorch · FastAPI · React · Three.js]
DeepCubeA algorithm: neural network + A* search to solve the Rubik's Cube. Interactive 3D visualization with Three.js.

French language model [Python · PyTorch]
Trained a transformer-based LLM on a French corpus using PyTorch.

MCP Drawing, real-time drawing via Claude Desktop [MCP · TypeScript · React · Express · Node.js]
MCP server in TypeScript with a React frontend for real-time canvas visualization through Claude Desktop.

TECHNICAL SKILLS

Languages: Python, Java, Haskell, TypeScript, JavaScript, C, C++, C#, PHP, Bash
AI / ML: PyTorch, TensorFlow, scikit-learn, LangChain, NumPy, Pandas, Matplotlib, OpenCV, YOLO, Gymnasium
Web / API: React, FastAPI, Django, Symfony, Flask, Node.js, Tailwind CSS, Three.js
Tools / DevOps: Docker, Terraform, Git, GitHub/GitLab, AWS, Scaleway, MCP, Jupyter, VS Code, JetBrains
Databases: PostgreSQL, MySQL, MongoDB, Neo4j, SQLite, SQL Server

LANGUAGES AND INTERESTS

Languages: French (native), English B2, Spanish A2
Interests: Formula 1, chess, running, collectible card games, myrmecology
Community: Kaggle competitions, published a Formula 1 dataset on Kaggle